

Disproof of Conjecture 1665: the proposed tight burning family

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Claim being disproved. The conjecture states, among other claims, that Cartesian products of the Petersen graph with paths have burning number $\lceil \sqrt{n} \rceil + 1$, where n is the order of the product. We disprove this universal tight-family statement. We make no assertion about the separate proposed upper bound for arbitrary graphs.

Definitions and graph. Let P be the Petersen graph on vertices $0, \dots, 9$. Its edges are the outer cycle $i(i+1)$ for $i \in \mathbb{Z}/5\mathbb{Z}$, the inner cycle $(5+i)(5+i+2)$, and the five spokes $i(5+i)$. Let P_2 be the path with two vertices $0, 1$. Set $G = P \square P_2$. Write the product vertex (u, j) as the integer $u + 10j$. Thus G is a simple graph of order 20.

At each burning round, choose a vertex unburned at the beginning of the round, ignite it, and spread the older fires to all adjacent vertices. For the burned set B and source s , the update is

$$B \mapsto B \cup \{s\} \cup \{v : (\exists u \in B) uv \in E(G)\}.$$

The burning number is the least number of rounds in which every vertex can be burned.

A four-round certificate. Choose the successive sources $0, 1, 2, 12$. Direct application of the update rule gives

$$\begin{aligned} B_0 &= \emptyset, \\ B_1 &= \{0\}, \\ B_2 &= \{0, 1, 4, 5, 10\}, \\ B_3 &= \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 15\}, \\ B_4 &= \{0, 1, \dots, 19\}. \end{aligned}$$

Each source is outside the burned set at the beginning of its round: $0 \notin B_0$, $1 \notin B_1$, $2 \notin B_2$, and $12 \notin B_3$. Consequently $b(G) \leq 4$. Equivalently, the Petersen graph has diameter two and its product with P_2 has diameter three, so one initial fire already reaches every vertex by the fourth round.

On the other hand, $4^2 < 20 \leq 5^2$, so $\lceil \sqrt{20} \rceil + 1 = 6$. Therefore

$$b(P \square P_2) \leq 4 < 6 = \lceil \sqrt{|V(P \square P_2)|} \rceil + 1.$$

This contradicts the claimed equality and disproves Conjecture 1665.

Lean verification and semantic correspondence. The self-contained file `lean/Main.lean` imports only `Std`. The predicate `petPath m` is the Cartesian product with the m -vertex path. Theorems `product_encoding` and `product_encoding_bijective` verify that the 20 labels used here represent exactly $P \square P_2$, and `graph_simple` checks symmetry and absence of loops. The definitions `advance`, `run`, and `legal` implement precisely the update rule above, including the condition on new sources. Theorems `legal_sources` and `all_burned` check the complete finite process, not merely its length. The predicate `BurningNumberIs` asserts successful burning and minimality. Finally, `not_tight_family_claim` refutes the universally quantified equality for all products with nonempty paths. Its reported axiom dependency is only `propext`. No `sorry`, `admit`, `native_decide`, or additional axiom is used.

Reproduction. Run `lean Main.lean` with Lean 4.19.0. The command exits successfully and prints the axiom dependencies of the principal results. All finite certificates use kernel-checked `decide`.